nature portfolio

Peer Review File

Machine learning identifies clusters of multimorbidity among decedents with inflammatory bowel disease

Corresponding Author: Dr Eric Benchimol

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thank you for the opportunity to review the manuscript by Postill et al. titled “Machine learning analyses multimorbidity among persons with inflammatory bowel diseases: a population-based study”

General comments
The manuscript explores multimorbidity among patients with inflammatory bowel diseases (IBD), tackling an important and timely matter given the increasing prevalence of IBD and the difficulties in managing individuals with multiple chronic health issues. The authors performed a retrospective analysis using data from Ontario, Canada, to investigate the clustering of chronic conditions in IBD patients. Their aim is to determine how multimorbidity accumulates over the life course of individuals with IBD. However, due to data limitations, the current iteration of the study does not fully meet its objectives.

I recommend that the authors reconsider their study objectives and contributions to research. It is important to note that the dataset lacks information on the onset dates of diseases, and the lookback period for diagnosing chronic conditions extends only to 1991 for individuals who passed away between 2010 and 2020. Consequently, the data reflect disease statuses for less than 30 years for those who died in that timeframe, which is insufficient for effectively determining "multimorbidity accumulation across the life course." Additionally, it is unclear how multimorbidity clustering patterns differ for individuals with IBD who share the same birth year as those included in the study but who did not die between 2010 and 2020.

The authors also compared the first identified disease among patients with IBD to those without IBD and found that the three most common non-IBD conditions identified first were hypertension, osteoarthritis and other arthritis, and mood disorders—conditions that also rank among the top diseases identified in the general population. However, it remains unclear how the disease accumulation sequences (i.e., the order of the first, second, and third diseases as recorded in the data) differ between patients with IBD and those without. I would like to know whether the authors were able to investigate how the sequences of disease accumulation differ between patients with IBD and those without IBD.

Below are some specific comments regarding the data, methods, and discussion sections.

Data
1. Research(e.g., Noble et all 2023) shows that inflammatory bowel disease (IBD) is strongly linked to ethnicity, lifestyle, and environmental factors. However, this study did not consider these variables. The authors should discuss the implications of this oversight on the matching results.
Noble AJ, Nowak JK, Adams AT, Uhlig HH, Satsangi J. Defining Interactions Between the Genome, Epigenome, and the Environment in Inflammatory Bowel Disease: Progress and Prospects. Gastroenterology. 2023 Jul;165(1):44-60.e2.
2. In lines 358-360, the authors note excluding cases with "invalid health cards, missing age, missing date of death, or those who were non-Ontario residents at the time of death." However, Supplemental Figure S1 indicates that only those with OHIP were excluded. Could the authors clarify this discrepancy?

Methods
3. The authors included the age at which diseases were identified (as recorded in the data) in their clustering analysis. I suggest that the authors conduct a sensitivity analysis excluding the disease-identified age. This will help to determine how the inclusion of this variable affects the clustering results and their interpretations, especially since the chronology captured

reflects the use of health services rather than the onset of the disease.

4. In Lines 365-367, the authors compared the first diagnosed disease in patients with inflammatory bowel disease (IBD) to those without IBD using matching methods. They included the year of death in their matching criteria. However, since the year of death is a post-exposure variable (following the diagnosis of IBD), and IBD may influence mortality, including the year of death in the matching process could introduce bias in the comparison.

5. The author created a correlation matrix of chronic conditions among individuals with IBD to identify highly correlated chronic conditions. However, the rationale for this step is unclear, as well as how it impacts the clustering.

6. Figures 6a and 6b for the deviation and validation cohorts appear identical. Could the author provide clarification on this?

Discussions
7. The study did not cluster diseases (e.g., Han et al. 2024); instead, it clustered populations. I recommend that the authors clarify this, as the current interpretation of the clustering results may be misleading. For example, the alpha-cluster, beta-cluster, and gamma-cluster represent three distinct populations rather than three separate clusters of diseases.
References: Han, S., Li, S., Yang, Y. et al. Mapping multimorbidity progression among 190 diseases. Commun Med 4, 139 (2024). https://doi.org/10.1038/s43856-024-00563-2
8. In Lines 253-293, the explanation of the current study should be more cautious. For example, the authors stated, "α-cluster reflects severe IBD symptoms, arising secondary to elevated systemic inflammation," which lacks direct evidence.

Reviewer #2

(Remarks to the Author)
Thank you for the opportunity to review this manuscript. This is a well-conducted study that employs advanced methods to compare multimorbidity between individuals with and without IBD. Overall, the study design is clear, and the methodology appears sound. I have a few comments and suggestions below:

Introduction
• When discussing multimorbidity as a growing focus, consider including statistical data on its prevalence or frequency in the populations of interest.
• It may improve the flow to introduce IBD first, followed by multimorbidity and common comorbidities among individuals with IBD. Providing statistics on IBD prevalence in Canada and globally could further strengthen this section.
• Consider adding a brief explanation (1–2 sentences) of how cluster analysis works for readers unfamiliar with machine learning and clustering methods.
• The sentence “Our findings inform multidisciplinary care needs for those with IBD.” seems more appropriate for the discussion or conclusion section, as it relates to the study’s impact rather than the background.

Methods
• Lines 136–137: It is unclear whether the timeframe (2010–2020) refers to the period of IBD diagnosis or the occurrence of death. Clarifying this would improve readability.
• A brief definition, such as “Clustering analysis in unsupervised machine learning is a technique that automatically groups similar data points based on their inherent patterns and relationships without predefined labels (…)”, could help readers unfamiliar with this advanced method. A figure visualizing the clustering process might also be useful.
• What were the rationales for choosing XGBoost? Providing justification for this choice would strengthen the methodological transparency.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thank you for the opportunity to review the revised manuscript by Postill et al. titled “Machine learning analyses of multimorbidity accumulation among persons with inflammatory bowel diseases: a population-based study”. Below are my specific comments:

1. The previous general comment: I encourage the authors to reevaluate their study objectives and contributions to the research. It's important to note that the study focuses on eligible descendants with IBD, rather than including persons with IBD, regardless of mortality status. To ensure clarity, I suggest modifying the title to: “…. among decedents with inflammatory bowel diseases….”

2. The previous general comment: The abstract and various sections of the revised manuscript claim that machine learning identifies "clusters of multimorbidity accumulation," implying temporal sequences of disease development. However, results fail to show such sequence-based clusters. This mismatch between methodology framing and actual findings creates conceptual confusion and risks misleading readers about the study’s contributions.

3. The previous general comment: The authors continue to frame their contribution using a “life-course” lens. However, life-course epidemiology typically examines exposures across the lifespan, particularly early-life factors (per their cited references). Given the truncated observation window (≤30 years) and absence of early-life data, this framing remains problematic. While methodological clarifications and sensitivity analyses are valuable, they do not resolve the core limitation of incomplete life-course data. I reiterate my suggestion to refocus the paper’s narrative to align with the available data.

4. The previous general comment: The analysis of disease accumulation sequences between IBD and non-IBD cohorts is a welcome addition. To strengthen the study’s objectives, I suggest explicitly comparing cluster patterns among decedents with IBD versus decedents without IBD. This would clarify whether observed patterns are uniquely associated with IBD in this population.

5. The previous comment 1: The claim that findings “can guide multidisciplinary care” requires further justification. Without accounting for ethnicity, lifestyle, or environmental factors—key determinants of multimorbidity—clinical applicability is unclear. Please temper interpretations and acknowledge this limitation in the Discussion, particularly regarding practical implications.

6. The previous comment 2: Please follow the relevant guidelines (e.g., STROBE) to report the study. For example, the authors need to clearly describe the sample selection criteria in the Study population from the Methods section to ensure reproducibility.

7. The previous comment 3: Given that “age at disease identification” reflects healthcare access rather than true onset, its inclusion in clustering may bias results. I continue to recommend conducting a sensitivity analysis that excludes this variable to evaluate its impact on cluster formation and interpretation. This is critical to support the authors’ assertion that diagnosis age is integral to their model.

8. The previous comment 6: Please carefully revise the mistakes in the figures. The numbers in Figures 7a and 7b for the original Figures 6), the deviation and validation cohorts are still identical in the manuscript.

9. Minor: “We excluded individuals with an invalid 424 health card, missing age, and/or missing date of death,….” Please remove “and/” to avoid logical ambiguity.

Reviewer #2

(Remarks to the Author)
Thank you for your further clarification. I do not have any other comments to add. Great works!

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thank you for your further clarification. I don't have any further comments to add.

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11-May-2025

Dear Dr. Barnes and members of the *Communications Medicine* Editorial Board,

Thank you for reviewing our manuscript entitled: "**Machine Learning Analyses of Multimorbidity among Persons with Inflammatory Bowel Disease: a Population-Based Study"** (COMMSMED-24-1476-T), now retitled "**Machine Learning Analyses of Multimorbidity Accumulation among Persons with Inflammatory Bowel Disease: A Population-Based Study"**.

Below we detail our response to reviewers and the line-by-line changes made in response to the comments. For clarity, we note the reviewer comments in italics with blue font color and our responses in black font color.

We hope these responses and revisions to our manuscript address most of the peer reviewers' questions and concerns. We look forward to your evaluation of this revised manuscript.

Sincerely,

Eric I. Benchimol, MD, PhD, FRCPC
Northbridge Chair in Inflammatory Bowel Disease, SickKids Inflammatory Bowel Disease Centre, The Hospital for Sick Children
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Reviewer #1 (Remarks to the Author):
Thank you for the opportunity to review the manuscript by Postill et al. titled "Machine learning analyses multimorbidity among persons with inflammatory bowel diseases: a population-based study"

General comments
The manuscript explores multimorbidity among patients with inflammatory bowel diseases (IBD), tackling an important and timely matter given the increasing prevalence of IBD and the difficulties in managing individuals with multiple chronic health issues. The authors performed a retrospective analysis using data from Ontario, Canada, to investigate the clustering of chronic conditions in IBD patients. Their aim is to determine how multimorbidity accumulates over the life course of individuals with IBD. However, due to data limitations, the current iteration of the study does not fully meet its objectives. I recommend that the authors reconsider their study objectives and contributions to research.

RESPONSE: Thank you for your review of our manuscript. We have revised the manuscript to address both your general and specific comments. We hope that these changes sufficiently address the comments provided. We update the study to read: *Machine Learning Analyses of Multimorbidity Accumulation among Persons with Inflammatory Bowel Disease: A Population-Based Study*. With respect to the study objective, we have also revised it for greater clarity; it now reads: "*Our overarching objective was to understand multimorbidity accumulation until time of death among those with IBD using a life course perspective.*" [Lines: 118-120]

It is important to note that the dataset lacks information on the onset dates of diseases, and the lookback period for diagnosing chronic conditions extends only to 1991 for individuals who passed away between 2010 and 2020. Consequently, the data reflect disease statuses for less than 30 years for those who died in that timeframe, which is insufficient for effectively determining "multimorbidity accumulation across the life course."

RESPONSE: We capture all incident diagnoses after 1990, as this is when the administrative health data in Ontario became digitized. Conditions with onset prior to this period are still identified through continued care for chronic conditions (continued care still triggers identification of the algorithm for diagnosis), as is the case for most conditions and patients included in this study. Therefore, it is unlikely that conditions were missed. We clarified this lookback period in our methods ("*The lookback window was from 1991 (date of earliest data) to death*." [Line: 360]) and specified this in the limitations:

> "*Secondly, the lookback windows for diagnoses of chronic conditions extend to 1991, and thus, it is possible the study may underestimate the burden of multimorbidity related to conditions diagnosed before this date. However, individuals with chronic health conditions diagnosed prior to 1991 would have received ongoing care for those conditions; these diagnoses would thus have been captured during the lookback window."* [Lines 296-300]

We recognize that the data is imperfect with respect to identifying multimorbidity accumulation; however, it is the best-source of population-level insight on multimorbidity care utilization available to us. Indeed, the administrative data used comes from a single payer system, capturing all healthcare interactions of >99% of the population of Ontario, Canada. Due to the length of available lookback window, we believe we capture the vast majority of chronic conditions that accumulate across the life course and model the care received for this multimorbidity within the last 30 years. We now reference the perspective piece by Wagner et al. 2024 (Link), to emphasize our definition of life course: *"We employ a life course perspective,[26] evaluating health patterns longitudinally to* *capture multimorbidity accumulation until time of death among people with IBD and their matched non-IBD decedents"* [Lines 126-128]

To more directly address your comment, clarify the implications of use of this data, and provide valuable insight to the readers, we have conducted a sensitivity analysis evaluating individuals born in 1960 or later (we detail this in the comment two down from this response) :

> *"As a sensitivity analysis, to evaluate the impact of administrative health data being digitized in 1991, we also evaluated the first three conditions of those born in 1960 and onwards as they would be $\leq$30 years old when digitized health administrative data became collected; their multimorbidity accumulation may thus be captured with greater sequential accuracy. However, these individuals' trajectories are unlikely to be reflective of the general population as they all die at <60 years of age*." [Lines: 448-454]

Additionally, it is unclear how multimorbidity clustering patterns differ for individuals with IBD who share the same birth year as those included in the study but who did not die between 2010 and 2020.

RESPONSE: We agree this is an interesting area of future research where we would examine a cohort of those living with IBD, but this is not within the scope of our study, as we included only those who died between 2010-2020. We clarified our objectives on this point: "*Our overarching objective was to understand multimorbidity accumulation until death among those with IBD using a life course perspective*." [Lines: 118-120]

We have also noted the need for such research moving forward: "*Firstly, we only include decedents, allowing a life course perspective of all conditions that accumulate until time of death; additional analyses with a living cohort can provide insights on how multimorbidity clustering among these individuals compares to that identified in our study."* [Lines: 293-295]

The authors also compared the first identified disease among patients with IBD to those without IBD and found that the three most common non-IBD conditions identified first were hypertension, osteoarthritis and other arthritis, and mood disorders—conditions that also rank among the top diseases identified in the general population. However, it remains unclear how the disease accumulation sequences (i.e., the order of the first, second, and third diseases as recorded in the data) differ between patients with IBD and those without. I would like to know whether the authors were able to investigate how the sequences of disease accumulation differ between patients with IBD and those without IBD.

RESPONSE: This is an interesting question and one of potential clinical value. We have revised Figure 1 to display an alluvial plot of disease accumulation sequences to provide the requested information:

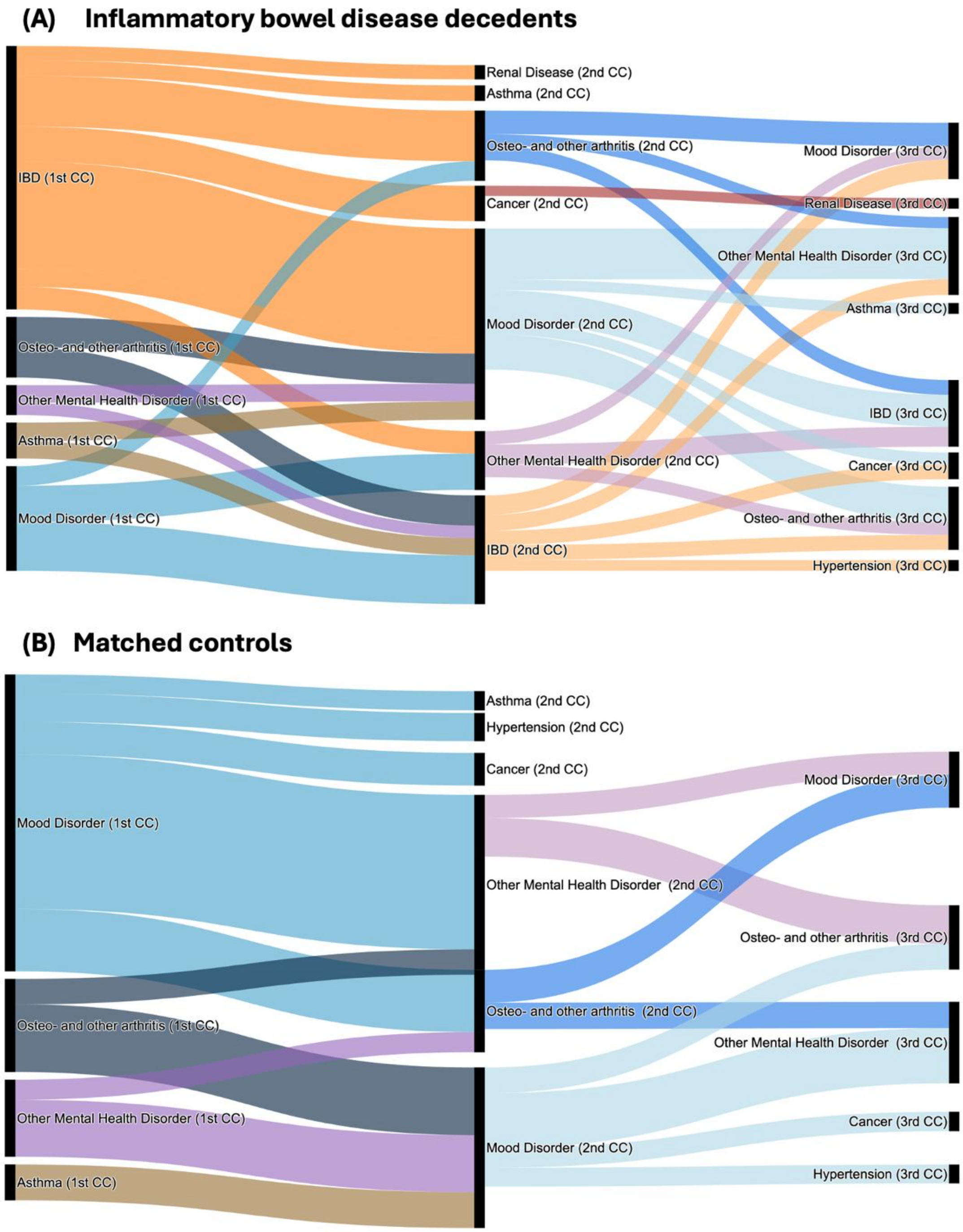


Figure 1. Alluvial plot of the most common pathways of chronic condition diagnosis/care as reported in the administrative health database for (A) decedents with IBD (N=9,278) and (B) matched non-IBD decedents (N=46,389). Note that the conditions displayed are not necessarily the most common conditions, but rather the most common sequence of condition acquisition. The thickness of the ribbons reflects the number of participants with which condition acquisition sequence in the data.
Figure Acronyms. IBD: Inflammatory Bowel Disease, CC: Chronic Condition

The methods section has been revised to include this figure: "*Additionally, we used alluvial plots to visualize common pathways of condition accumulation.*" [Lines: 448]

The results of this are displayed in the **Supplemental Table S2**. We attach **Supplemental Table S2** at the bottom of this document for ease of review.

Below are some specific comments regarding the data, methods, and discussion sections.

RESPONSE: Thank you for these comments. We address them line-by-line below.

Data
R1.1. Research(e.g., Noble et all 2023) shows that inflammatory bowel disease (IBD) is strongly linked to ethnicity, lifestyle, and environmental factors. However, this study did not consider these variables. The authors should discuss the implications of this oversight on the matching results.
Noble AJ, Nowak JK, Adams AT, Uhlig HH, Satsangi J. Defining Interactions Between the Genome, Epigenome, and the Environment in Inflammatory Bowel Disease: Progress and Prospects. Gastroenterology. 2023 Jul;165(1):44-60.e2.

RESPONSE: There are important differences in ethnicity, lifestyle, and environmental factors among those with and without IBD. These factors both influence and are influenced by IBD. Our objective was to assess the burden of multimorbidity among those with IBD at death and compare its occurrence to matched controls. We intentionally did not match on ethnic and lifestyle factors as these factors also influence the occurrence of multimorbidity. We now reference Kaufman to provide further rationale of our choice:

> Kaufman JS. Statistics, Adjusted Statistics, and Maladjusted Statistics. Am J Law Med. 2017 May;43(2-3):193-208. doi: 10.1177/0098858817723659. PMID: 29254468.

Our approach was not causal, but rather descriptive, observing how conditions co-occur across groups without removing differences in the distribution of ethnic, lifestyle, and environmental factors across groups. We clarify this and include the reference provided (Noble et al. 2023) in the limitations section:

> *"Fourthly, we compare to matched controls (vs. the entire general population), to ensure comparable lookback windows. To improve the representation of matched controls to the general population, we did not match decedents with IBD and those without IBD on ethnic, lifestyle, and environmental factors – though these factors differ between groups[48] and can influence the occurrence of multimorbidity. Our study sought to generate population-insights that can guide multidisciplinary care without erasing differences in the distribution of ethnic, lifestyle, and environmental factors across groups.[52] Despite this, we observed area-level socioeconomic measures (based on decedent's end of life postal code) to be balanced across groups when matched on year of birth and year of death." (**Supplemental Table S5**)."* [Line: 305-313]

In Table 1 we show that decedents with and without IBD were well balanced on area-level material deprivation, education, and income. In this revision, we have also added a measure of rurality to Table 1 to enable comparison across decedents with and without IBD. Our measure of rurality (Rurality Index of Ontario) is described in the methods: "*Rurality was defined using the Rurality Index of Ontario scores, with scores $\geq$40 indicating rural.[71]*"[Line 411-412]

R1.2. In lines 358-360, the authors note excluding cases with "invalid health cards, missing age, missing date of death, or those who were non-Ontario residents at the time of death." However, Supplemental Figure S1 indicates that only those with OHIP were excluded. Could the authors clarify this discrepancy?

RESPONSE: We have added a figure caption to **Supplemental Figure S1** for improved clarity on this point:

> *Notes*. We excluded those with invalid health cards, missing age at death, and/or missing date of death; we also excluded non-Ontario residents at the time of death (ineligible for OHIP). Having either an invalid health card, missing age at death, and/or missing date of death meant the individual did not meet study inclusion definition (excluded prior to creating the potential study population); these individuals are not shown in the figure as they were not present in the initial dataset. We then excluded individuals who were ineligible for OHIP at death.

Methods

R1.3. The authors included the age at which diseases were identified (as recorded in the data) in their clustering analysis. I suggest that the authors conduct a sensitivity analysis excluding the disease-identified age. This will help to determine how the inclusion of this variable affects the clustering results and their interpretations, especially since the chronology captured reflects the use of health services rather than the onset of the disease.

RESPONSE: While we find this a particularly interesting question, the appropriate clustering approach is dependent on the data size and type inputted into the clustering models. We used Consensus K-Means given our mid-sized cohort and the input of only numerical values (age of diagnosis) into the cluster. If we used only binary indicators of presence/absence of conditions, it would also be pertinent to re-evaluate the appropriate clustering model (e.g., likely K-Modes would be appropriate). We see the use of an additional clustering model to cluster a different format of data as beyond the scope of this project. Using age at diagnosis provides us with more information by including not just the presence or absence of the conditions, but also their timing which may be important in determining the disease profiles. Elsewhere in the literature, there is also a shift towards including temporal data in multimorbidity clustering:

> Holm NN, Le TM, Frølich A, Andersen O, Juul-Larsen HG, Stockmarr A, Venkatesh S. amVAE: Age-aware multimorbidity clustering using variational autoencoders. Comput Biol Med. 2025 Mar;186:109632. doi: 10.1016/j.compbiomed.2024.109632. Epub 2025 Jan 16. PMID: 39823822.

We agree that health service use does not indicate *biological* disease onset. However, we note that health service use is required for a patient to receive a diagnosis of a disease from a healthcare professional. The algorithms we used to identify the timepoints at which a patient had a diagnosis were identified through validated algorithms. We provide details on the sensitivity and specificity as well as the billing codes used to identify these conditions in Supplemental Table S1**.**

R1.4. In Lines 365-367, the authors compared the first diagnosed disease in patients with inflammatory bowel disease (IBD) to those without IBD using matching methods. They included the year of death in their matching criteria. However, since the year of death is a post-exposure variable (following the diagnosis of IBD), and IBD may influence mortality, including the year of death in the matching process could introduce bias in the comparison.

RESPONSE: Indeed, those with IBD have shorter life expectancies than those without. Our study is descriptive, comparing how condition co-occur across the life course among those who died with and without IBD. Given the constraints of the data noted above, we believe that ensuring those with and without IBD had comparative periods of data collection during their life course was necessary to prevent information bias. To do so, required matching on both year of birth and year of death. We now explain this in the methods: "*We matched IBD and non-IBD decedents on both year of birth and year of death to ensure we had comparable windows of data collection for both groups.*" [Line 370-371]

We have also revised to provide further explanation and clarity on this in in the discussion:

> *"Fourthly, we compare to matched controls (vs. the entire general population), to ensure comparable lookback windows. To improve the representation of matched controls to the general population, we did not match decedents with IBD and those without IBD on ethnic, lifestyle, and environmental factors – though these factors differ between groups[48] and can influence the occurrence of multimorbidity. Our study sought to generate population-insights that can guide multidisciplinary care without erasing differences in the distribution of ethnic, lifestyle, and environmental factors across groups.[52] Despite this, we observed area-level socioeconomic measures (based on decedent's end of life postal code) to be balanced across groups when matched on year of birth and year of death." (**Supplemental Table S5**)."* [Line: 305-313]

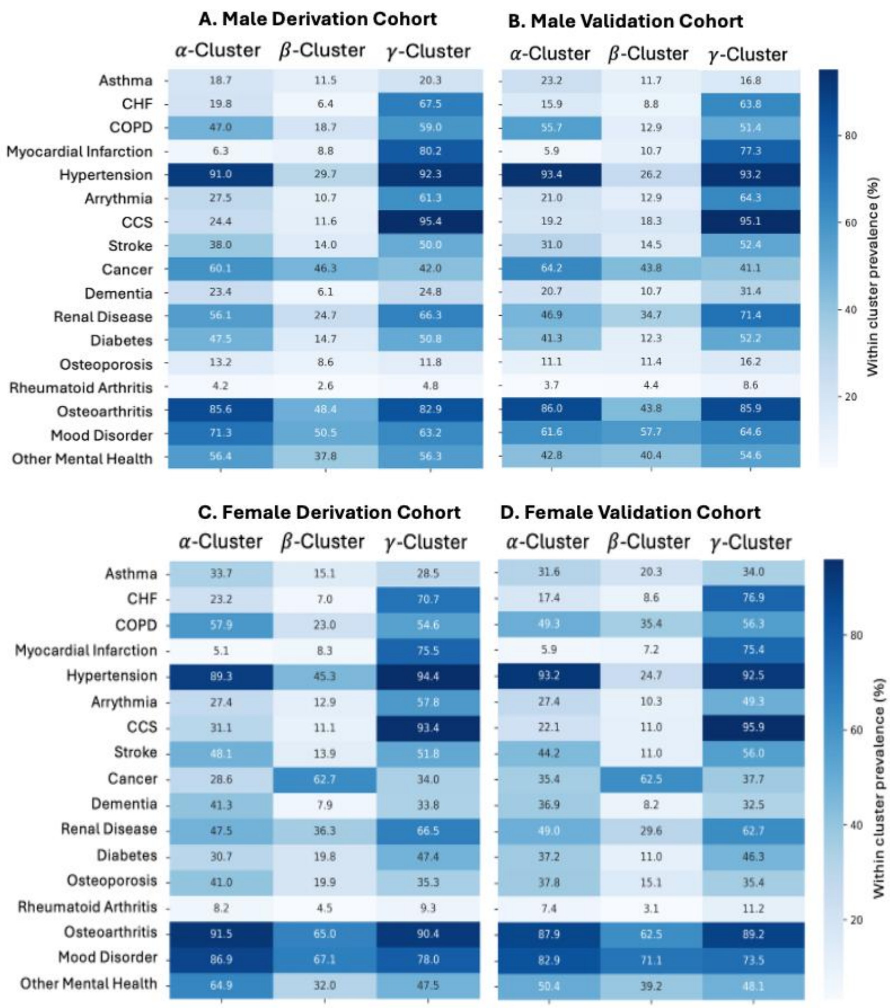
R1.5. The author created a correlation matrix of chronic conditions among individuals with IBD to identify highly correlated chronic conditions. However, the rationale for this step is unclear, as well as how it impacts the clustering.

RESPONSE: For improved clarity, we have added the following to the supplemental file

> "*We created a correlation matrix as best practice for initial data exploration. Evaluating feature correlation can also be important for interpreting cluster performance, particularly with linear clustering methods (e.g. K-Means). In this study K-Means with Euclidean distance as our distance metric was the base model for our consensus clustering approach. K-Means assumes spherical clusters; including correlated features can elongate clusters along the correlated dimension. Correlated features can also add instability if correlated features dominate clustering processes; in such case small (non-clinically meaningful) variability in such features may shift clusters centroids or results in clusters with differences that are not clinically meaningful. Finally, correlated features increase dimensionality (and computational cost) without adding much new information. We prespecified a threshold of ≥0.8 for removal of highly correlated features*." [Page 10].

R1.6. Figures 6a and 6b for the deviation and validation cohorts appear identical. Could the author provide clarification on this?

RESPONSE: Thank you for noting this. The identical values were an error (the values from the duplicated cohort were inputted twice.). Below is the correct version of Figure 6.



Discussions

R1.7. The study did not cluster diseases (e.g., Han et al. 2024); instead, it clustered populations. I recommend that the authors clarify this, as the current interpretation of the clustering results may be misleading. For example, the alpha-cluster, beta-cluster, and gamma-cluster represent three distinct populations rather than three separate clusters of diseases.

References: Han, S., Li, S., Yang, Y. et al. Mapping multimorbidity progression among 190 diseases. Commun Med 4, 139 (2024). https://doi.org/10.1038/s43856-024-00563-2

RESPONSE: This is correct, we are clustering the multimorbidity healthcare utilization patterns of decedents, all of whom had IBD at death. We have clarified the introduction and objective to:

- [Lines: 110-112] "*To properly provide holistic patient-centered care for those with IBD, multimorbidity research in the IBD patient population is needed. Yet, to date, no work has examined population-level patterns of multimorbidity care utilization across the life course in the IBD population.*"
- [Lines: 118-120] "*Our overarching objective was to understand multimorbidity accumulation until death among those with IBD using a life course perspective.*"

We have also revised elements of the discussion so that this is clearer. Specifically, we modified:

- [Lines: 223-224] *"Individuals with IBD had distinct data-driven patterns of care utilization for multimorbidity that formed three stable and reproducible clusters. Such results are pertinent to informing the allocation of resources for those with IBD at the population level."*

- [Lines 256-261] *"Decedents with IBD had distinct multimorbidity care patterns: (α-Cluster) IBD with hypertension, mood disorder, other mental health disorders and/or osteo- and other arthritis, (β-Cluster) IBD with high prevalence of cancer and low multimorbidity, and (γ-Cluster) IBD with mood disorders and cardiovascular comorbidities. A unique aspect of our approach is that we used data reflective of the order and timing with which individuals sought care for their chronic conditions, irrespective of their IBD severity."*

R1.8. In Lines 253-293, the explanation of the current study should be more cautious. For example, the authors stated, "α-cluster reflects severe IBD symptoms, arising secondary to elevated systemic inflammation," which lacks direct evidence.

RESPONSE: We agree that there lacks direct evidence for this claim. The original sentence began with: "*We postulate that the α-Cluster reflects severe IBD symptoms…* " Our speculation is based on prior literature and the findings of our study. We have modified to make clearer that this is speculation, rather than evidence from our study:

> "*A unique aspect of our approach is that we used data reflective of the order and timing with which individuals sought care for their chronic conditions, irrespective of their IBD severity. However, it is possible that these clusters reflect elements of IBD severity. For example, conditions in the α-Cluster could arise secondary to elevated systemic inflammation and/or chronic pain, given the relationships explained above."* [Lines 259-263]

Reviewer #2 (Remarks to the Author):

Thank you for the opportunity to review this manuscript. This is a well-conducted study that employs advanced methods to compare multimorbidity between individuals with and without IBD. Overall, the study design is clear, and the methodology appears sound. I have a few comments and suggestions below:

RESPONSE: We thank you for the positive review of our manuscript and these helpful comments.

Introduction

R2.1. When discussing multimorbidity as a growing focus, consider including statistical data on its prevalence or frequency in the populations of interest.

RESPONSE: We have revised the introduction to include:

> "*In Ontario, Canada, the prevalence of multimorbidity at death in the general population rose from 80% in 1994 to 95% in 2013,*[7] *with prevalence increases greatest among younger ages.*[8] *Multimorbidity is an area of increasing focus in medical and public health research, given its increasing prevalence*[7,9] *and association with more complex care plans,*[10] *increased healthcare utilization and costs,*[11] *excess mortality,*[12,13] *and poorer health outcomes.*[14] *Multimorbidity is not solely a problem of aging, as it actually confers greater risk of mortality in early and middle adulthood.*[13]*"* [Lines: 103-110]

R2.2. It may improve the flow to introduce IBD first, followed by multimorbidity and common comorbidities among individuals with IBD. Providing statistics on IBD prevalence in Canada and globally could further strengthen this section.

RESPONSE: Thank you for this suggestion, we have revised the ordering of the introduction to begin with IBD. We also added the following detail on the prevalence of IBD: "*Canada has among the highest incidence and prevalence of IBD in the world.*[3] *Due to advances in treatment and prognosis of IBD, patients are living longer* [2] *and thus, multimorbidity is an emerging concept of study in IBD.*[4] *Indeed, the prevalence of IBD is continuing to grow rapidly, such that approximately 470,000 (equivalent to 1 in 91) Canadians are forecasted to be living with IBD by 2035.*[5] " [Lines: 95-101]

R2.3. Consider adding a brief explanation (1–2 sentences) of how cluster analysis works for readers unfamiliar with machine learning and clustering methods.

RESPONSE: We have provided additional details on clustering in the introduction: "*Machine learning (ML), the use of computer-based algorithms for flexibly identifying patterns within datasets, is increasingly used to understand multimorbidity.*[15–17]" [Lines: 114-115]

Additionally, when discussing the methodology, we re-clarify this: "*One ML approach to studying multimorbidity is clustering– a technique that automatically groups similar data points based on their inherent patterns and relationships without predefined labels.*" [Lines 115-117] We have also added a new figure (called Figure 4) to explain the clustering approach used as you recommend in a later comment.

R2.4. The sentence "Our findings inform multidisciplinary care needs for those with IBD." seems more appropriate for the discussion or conclusion section, as it relates to the study's impact rather than the background.

RESPONSE: We have removed this sentence.

Methods

R2.5. Lines 136–137: It is unclear whether the timeframe (2010–2020) refers to the period of IBD diagnosis or the occurrence of death. Clarifying this would improve readability.

RESPONSE: We have revised so that it reads: "*…decedents with IBD and those without IBD with dates of death between 2010-2020 in Ontario, Canada.*" [Lines 125-126]

R2.6. A brief definition, such as "Clustering analysis in unsupervised machine learning is a technique that automatically groups similar data points based on their inherent patterns and relationships without predefined labels (…)", could help readers unfamiliar with this advanced method. A figure visualizing the clustering process might also be useful.

RESPONSE: We have provided a more robust definition of clustering analysis: "*One ML approach to studying multimorbidity is clustering – a technique that automatically groups similar data points based on their inherent patterns and relationships without predefined labels.*" [Lines: 115-117] As suggested, we have added a new figure (called Figure 4) and an in-depth figure caption to explain the clustering approach used:

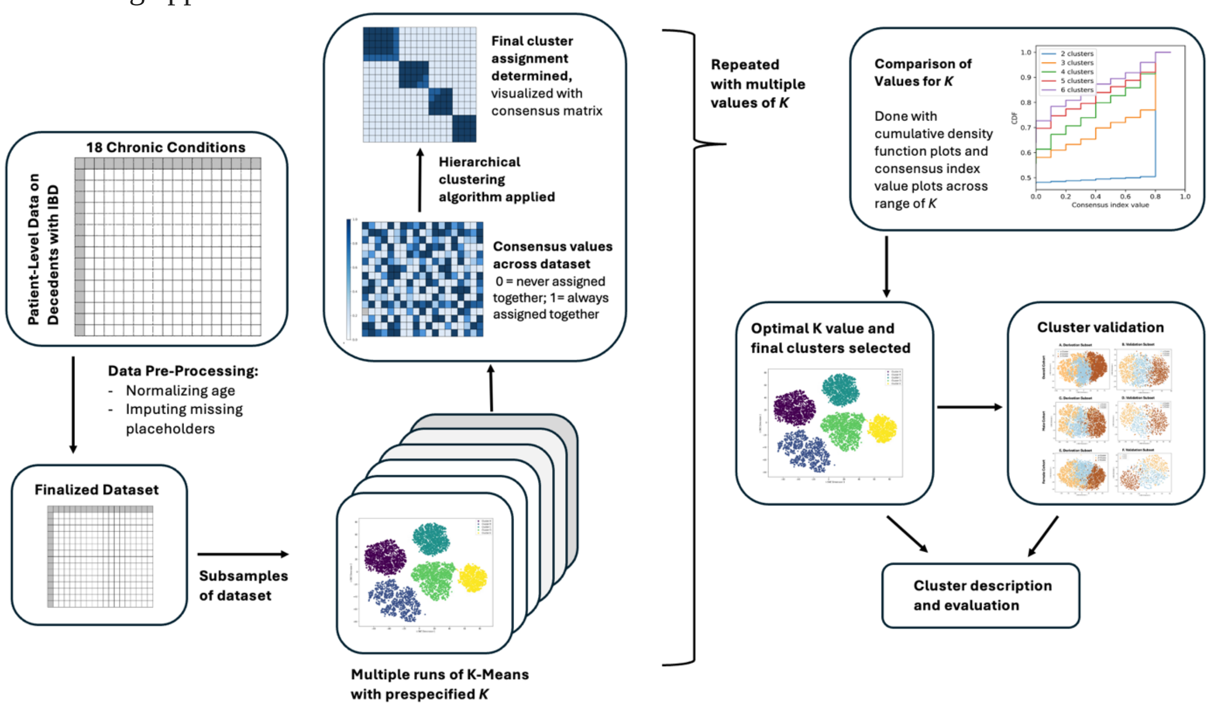


Figure 4. Clustering approach leveraged. Dataset containing age of diagnosis of 18 chronic conditions for decedents with IBD is pre-processed, which includes normalizing the age of diagnosis for each condition and imputing placeholders for those without age of diagnosis. Across the range of pre-determined values for *K* (K=2 to K=6), consensus clustering is done. Consensus clustering is a grouping approach that incorporates results from multiple runs of an inner-loop clustering algorithm on sub-sampled subjects. To generate the final clusters for a pre-specific value of *K* (number of clusters), the dataset is subsampled many times. The base clustering algorithm is run on each subsample of the data. Consensus scores are then generated to compare the stability across each clustering run. Consensus scores reflect the proportion of times that an individual data point is assigned to the same cluster in the subsamples (0 = never assigned to the same cluster; 1 = always assigned to the same cluster). A hierarchical clustering algorithm is used to generate the final clusters and consensus matrix. The quality of the final clusters generated for each value of K are compared. The final clusters are identified, validated, and described.

R2.7. What were the rationales for choosing XGBoost? Providing justification for this choice would strengthen the methodological transparency.

RESPONSE: Thank you for this suggestion. We have added the following justification for this choice to the manuscript:

> "*We selected XGBoost as the classification model for validating our unsupervised clustering results due to its explainability and strong predictive performance with other healthcare mortality prediction tasks,*[83–90] *ability to handle complex, non-linear interactions between features, regularization mechanisms that prevent overfitting, and efficient implementation of the gradient boosting framework. Additionally, XGBoost is amenable to multiclass prediction tasks as is required for predicting specific cluster membership (as opposed to a 1 vs. rest approach). By achieving high predictive accuracy with such algorithm, we can demonstrate that identified clusters were not arbitrary but reflective of distinguishable patterns in the data*." [Lines 530-537]

Supplemental Table S2. The first three chronic conditions diagnosed among decedents with inflammatory bowel disease and matched non-IBD decedents, as reported in the administrative health data.

	Overall Cohort			Subset of Cohort Born in 1960 or Later		
Chronic condition, n (%)	**IBD Decedents** (N=9,278)	**Matched Non-IBD Decedents** (N=45,923)	**SD**	**IBD Decedents** (N=1,264)	**Matched Non-IBD Decedents** (N=6,366)	**SD**
First Conditions Diagnosed						
Inflammatory Bowel Disease	3,172 (34.2)	0 (0.0)	-	618 (48.9)	0 (0.0)	-
Arrythmia	87 (0.9)	743 (1.6)	6.1%	101 (8.0)	560 (9.2)	4.5%
Asthma	428 (4.6)	2,210 (4.8)	1.0%%	1-5 (<0.3)	23 (0.4)	4.3%
Chronic Coronary Syndrome	260 (2.8)	1,801 (3.9)	6.2%	0 (0.0)	78 (1.3)	-
Congestive Heart Failure	35-39 (<0.5)	493 (1.1)	8.1%	1-5 (<0.3)	61 (1.0)	11.2%
COPD	288 (3.1)	2,153 (4.7)	8.2%	29 (2.3)	421 (7.0)	22.3%
Cancer	327 (3.5)	2,775 (6.0)	11.8%	1-5 (<0.3)	86 (1.4)	11.9%
Dementia	1-5 (<0.1)	216 (0.5)	-	1-5 (<0.3)	35 (0.6)	3.9%
Diabetes	225 (2.4)	2,679 (5.8)	17.2%	1-5 (<0.3)	79 (1.3)	13.5%
Hypertension	1,365 (14.7)	11,191 (24.4)	24.6%	12 (0.9)	462 (7.6)	33.4%
Mood disorder	1,128 (12.2)	7,720 (16.8)	13.3%	0 (0.0)	1-5 (0.1)	-
Myocardial Infarction	90 (1.0)	843 (1.8)	7.4%	9 (0.7)	63 (1.0)	3.5%
Osteo- and Other Arthritis	1,407 (15.2)	9,104 (19.8)	12.3%	15 (1.2)	277 (4.6)	20.4%
Osteoporosis	66 (0.7)	451 (1.0)	3.0%	1-5 (<0.3)	21 (0.3)	2.0%
Other Mental Health Disorder	227 (2.4)	2,177 (4.7)	12.4%	1-5 (<0.3)	24 (0.4)	6.5%
Renal Disease	49 (0.5)	348 (0.8)	2.9%	146 (11.6)	1037 (17.1)	16.0%
Rheumatoid Arthritis	47 (0.5)	250 (0.5)	0.5%	241 (19.1)	2065 (34.1)	34.5%
Stroke	74 (0.8)	718 (1.6)	7.1%	75 (5.9)	760 (12.5)	23.0%
Second Condition Diagnosed						
Inflammatory Bowel Disease	1,463 (15.8)	0 (0.0)	-	236 (19.0)	0 (0.0)	-
Arrythmia	171 (1.9)	1,181 (2.7)	5.4%	17 (1.4)	107 (2.0)	4.7%
Asthma	306 (3.3)	1,247 (2.8)	3.0%	55 (4.4)	217 (4.0)	2.2%
Chronic Coronary Syndrome	421 (4.6)	2,706 (6.1)	6.8%	10 (0.8)	80 (1.5)	6.3%
Congestive Heart Failure	82 (0.9)	799 (1.8)	7.9%	1-5 (<0.3)	56 (1.0)	8.7%
COPD	474 (5.1)	2,595 (5.8)	3.1%	16 (1.3)	143 (2.6)	9.7%
Cancer	301 (3.3)	2,630 (5.9)	12.7%	76 (6.1)	492 (9.1)	11.1%
Dementia	25 (0.3)	544 (1.2)	11.1%	1-5 (<0.3)	13 (0.2)	-
Diabetes	344 (3.7)	2,704 (6.1)	10.9%	21 (1.7)	218 (4.0)	14.0%
Hypertension	1,149 (12.4)	6,920 (15.5)	9.0%	52 (4.2)	395 (7.3)	13.3%
Mood disorder	1,557 (16.9)	6,819 (15.3)	4.2%	354 (28.6)	1228 (22.7)	13.6%
Myocardial Infarction	170 (1.8)	1,296 (2.9)	7.0%	14 (1.3)	143 (2.6)	9.7%
Osteo- and Other Arthritis	1,717 (18.6)	7,928 (17.8)	2.0%	170 (13.7)	780 (14.4)	1.9%
Osteoporosis	192 (2.1)	812 (1.8)	1.8%	16 (1.3)	28 (0.5)	8.2%
Other Mental Health Disorder	1,557 (16.9)	6,819 (15.3)	4.2%	165 (13.3)	676 (12.5)	2.5%
Renal Disease	160 (1.7)	904 (2.0)	2.2%	37 (3.0)	190 (3.5)	2.9%
Rheumatoid Arthritis	49 (0.5)	220 (0.5)	0.5%	5-10 (<0.8)	23 (0.4)	2.0%
Stroke	190 (2.1)	1,591 (3.6)	9.2%	17 (1.4)	145 (2.7)	9.3%
Third Conditions Diagnosed						
Inflammatory Bowel Disease	1,251 (13.8)	0 (0.0)	-	167 (14.4)	0 (0.0)	-
Arrythmia	233 (2.6)	1,673 (4.0)	8.1%	20 (1.7)	121 (2.8)	7.2%
Asthma	293 (3.2)	1,240 (3.0)	1.5%	54 (4.7)	194 (4.5)	0.9%
Chronic Coronary Syndrome	492 (5.4)	3,208 (7.7)	9.2%	15 (1.3)	122 (2.8)	10.7%

Congestive Heart Failure	131 (1.4)	1,214 (2.9)	10.1%	6 (0.5)	76 (1.8)	11.7%
COPD	498 (5.5)	2,740 (6.6)	4.6%	35 (3.0)	231 (5.3)	11.6%
Cancer	430 (4.7)	3,087 (7.4)	11.2%	103 (8.9)	478 (11.0)	7.2%
Dementia	52 (0.6)	9,60 (2.3)	14.6%	0 (0.0)	23 (0.5)	-
Diabetes	288 (3.2)	1,530 (3.7)	2.7%	36 (3.1)	224 (5.2)	10.4%
Hypertension	1,108 (12.2)	5,330 (12.8)	1.8%	36 (3.1)	224 (5.2)	10.4%
Mood disorder	1,165 (12.8)	4,433 (10.6)	6.8%	173 (14.9)	571 (13.2)	5.0%
Myocardial Infarction	237 (2.6)	1,761 (4.2)	8.9%	14 (1.2)	112 (2.6)	10.1%
Osteo- and Other Arthritis	1,318 (14.5)	5,438 (13.0)	4.3%	14 (1.2)	112 (2.6)	10.1%
Osteoporosis	270 (3.0)	940 (2.3)	4.5%	34 (2.9)	25 (0.6)	18.0%
Other Mental Health Disorder	598 (6.6)	3,177 (7.6)	4.0%	165 (14.2)	676 (15.6)	3.9%
Renal Disease	288 (3.2)	1,530 (3.7)	2.7%	67 (5.8)	284 (6.5)	3.2%
Rheumatoid Arthritis	64 (0.7)	222 (0.5)	2.2%	8 (0.7)	22 (0.5)	2.4%
Stroke	285 (3.1)	2,178 (5.2)	10.4%	24 (2.1)	181 (4.2)	12.1%

Note. Cells <5 are suppressed due to privacy regulations. Additional cells are suppressed to prevent back calculation where necessary.

Note. Not all decedents develop a second or third chronic condition and thus, column totals do not add to population totals.

Table Acronyms. IBD: Inflammatory Bowel Disease, COPD: Chronic Obstructive Pulmonary Disorder

We thank you for the opportunity to pursue a second round of revisions. We have provided substantial revisions to more fully address Reviewer #1's original comments. We have also reduced the word count of the manuscript to allow for the new sensitivity analysis suggested by Review #1.

Reviewer #1 (Remarks to the Author)

Thank you for the opportunity to review the revised manuscript by Postill et al. titled "Machine learning analyses of multimorbidity accumulation among persons with inflammatory bowel diseases: a population-based study". Below are my specific comments:

RESPONSE: We are very appreciative of your additional review of the manuscript and the helpful comments below. We have substantially revised the manuscript to address your comments as indicated below. Particularly we clarify focus on decedents with IBD and add the additional sensitivity analyses suggested. We provide a point-by-point response.

1. The previous general comment: I encourage the authors to reevaluate their study objectives and contributions to the research. It's important to note that the study focuses on eligible descendants with IBD, rather than including persons with IBD, regardless of mortality status. To ensure clarity, I suggest modifying the title to: "…. among decedents with inflammatory bowel diseases…."

RESPONSE: We agree. The title has been modified as you suggested. It now reads: "Machine Learning Identified Clusters of Multimorbidity among Decedents with Inflammatory Bowel Disease: A Population-Based Study"

To further clarify that we are describing multimorbidity prior to death, we have also revised **Figure 2** to focus on the prevalence of chronic conditions at death (we removed the panel displaying conditions at age 60 years as the prevalence here would not be reflective of that in a living population) and **Figure 3** to now display the accumulation of conditions in the 20 years prior to death (as opposed to across the life time).

Revised Figure 2:

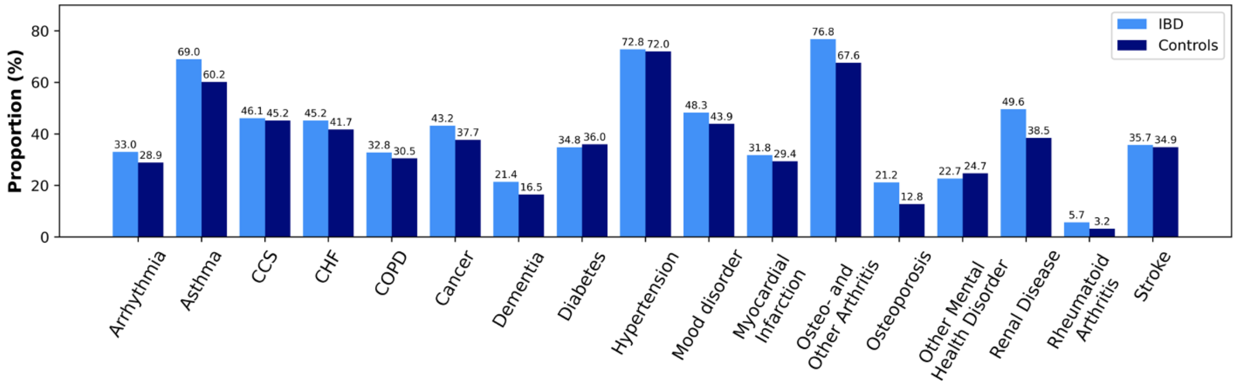


Figure 2. Prevalence of all chronic conditions at the time of death in people with inflammatory bowel disease (IBD, N=9,278) and matched non-IBD (N=46,389) decedents.

Revised Figure 3:

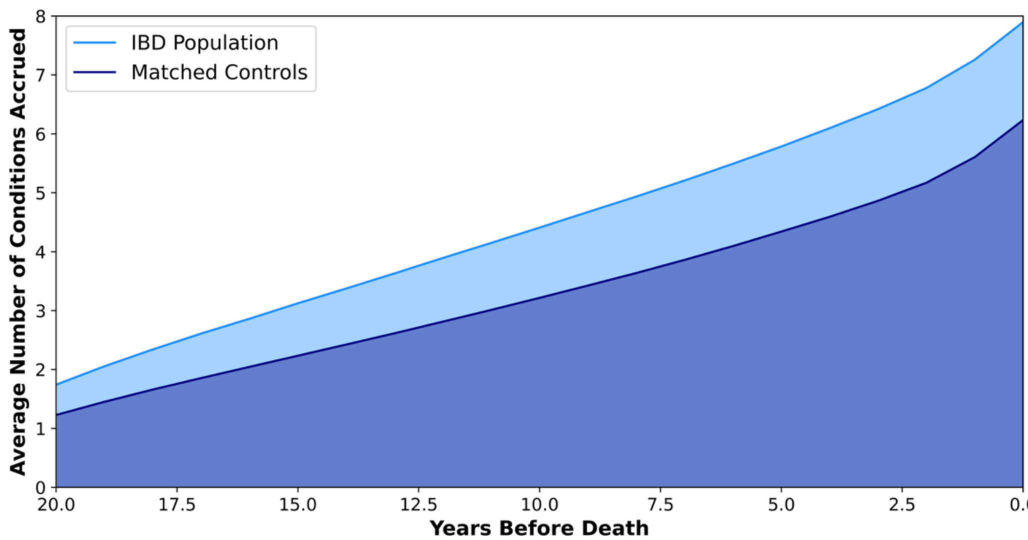


Figure 3. Accumulation of chronic conditions prior to death among decedents with inflammatory bowel disease (IBD, N=9,278) compared to matched non-IBD decedents (N=46,389). Figure displays the average number of conditions those with and without IBD had in the years before their death (alpha = 0.4 smoothing applied).

2. The previous general comment: The abstract and various sections of the revised manuscript claim that machine learning identifies "clusters of multimorbidity accumulation," implying temporal sequences of disease development. However, results fail to show such sequence-based clusters. This mismatch between methodology framing and actual findings creates conceptual confusion and risks misleading readers about the study's contributions.

RESPONSE: We have revised the language for greater conceptual clarity:

- In the abstract, it now reads: "Our objective was to identify patterns of multimorbidity that occur prior to death among people with inflammatory bowel disease (IBD)."
- In the introduction, lines 119-120: "Our overarching objective was to understand multimorbidity accumulation that occurs before death among those with IBD."
- In the introduction, lines 128-130: "We then utilized unsupervised ML to identify multimorbidity clusters at death among those with IBD. Our clustering approach differs from prior approaches in that we incorporate the age of healthcare utilization for chronic conditions in the data used to derive the clusters."

3. The previous general comment: The authors continue to frame their contribution using a "life-course" lens. However, life-course epidemiology typically examines exposures across the lifespan, particularly early-life factors (per their cited references). Given the truncated observation window (≤30 years) and absence of early-life data, this framing remains problematic. While methodological clarifications and sensitivity analyses are valuable, they do not resolve the core limitation of incomplete life-course data. I reiterate my suggestion to refocus the paper's narrative to align with the available data.

RESPONSE: We have removed the wording "life-course" throughout the manuscript and instead reference our approach as providing insight on "multimorbidity prior to death." This is also reflected in the revised subheadings in the results section.

4. The previous general comment: The analysis of disease accumulation sequences between IBD and non-IBD cohorts is a welcome addition. To strengthen the study's objectives, I suggest explicitly comparing cluster patterns among decedents with IBD versus decedents without IBD. This would clarify whether observed patterns are uniquely associated with IBD in this population.

RESPONSE: We agree that this meaningfully augments the paper and robustness of our interpretation. We have re-run our analysis on the controls.

- We document this as a sensitivity analysis in the methods: "To clarify if the observed multimorbidity patterns are unique to IBD in this population, we repeated the clustering approach on the matched controls." [Lines 492 – 493]
- We document the results: "In the matched controls, the same clustering approach derived Cluster A (cancer), Cluster B (analogous to the IBD γ-Cluster), Cluster C (analogous to the IBD β-Cluster), and Cluster D (no cancer, hypertension, osteo-arthritis, and mood disorder), as further detailed in **Supplemental Table S12**." [Lines 493 – 495]
- We provide further detail in supplemental file. For ease of review, we have also added the supplemental figures and table for this sensitivity analysis to the bottom of this document.

5. The previous comment 1: The claim that findings "can guide multidisciplinary care" requires further justification. Without accounting for ethnicity, lifestyle, or environmental factors—key determinants of multimorbidity—clinical applicability is unclear. Please temper interpretations and acknowledge this limitation in the Discussion, particularly regarding practical implications.

RESPONSE: We have removed this claim and clarified this limitation as you suggested. It now reads:

> "Our study sought to generate population-insights without erasing differences in the distribution of ethnic, lifestyle, and environmental factors across groups.[53] However, with respect to clinical applicability, our findings should be considered along for ethnicity, lifestyle, or environmental factors, key determinants of multimorbidity." [Lines 300 – 303]

We also softened the language in the abstract: "These findings can inform future research efforts and potential multimorbidity care needs in populations with IBD. "

6. The previous comment 2: Please follow the relevant guidelines (e.g., STROBE) to report the study. For example, the authors need to clearly describe the sample selection criteria in the Study population from the Methods section to ensure reproducibility.

RESPONSE: We followed the appropriate extension to the STROBE guidelines for research using routinely collected health data (RECORD), which is also included in this re-submission. We had included details on sample selection criteria in the methods section as well as in the supplemental file (**Supplemental Figure S1**). We revised the wording for greater clarity:

> "Participants were all residents of Ontario whose deaths were dated between January 1, 2010, and January 31, 2020. The Ontario Crohn's and Colitis Cohort (OCCC) was used to identify residents of Ontario with IBD using validated age-specific algorithms […] Controls were identified from the Registered Persons Database (RPDB), a central population registry file that contains basic demographic information such as birth date, sex, and postal code for those who have received an Ontario health card number for the province's universal health care system since 1992. We matched eligible decedents with IBD to up to 5 decedents without IBD from the general population at random, based on sex, year of birth, and year of death, using an exact match approach without replacement." [Lines 341 – 357]

7. The previous comment 3: Given that "age at disease identification" reflects healthcare access rather than true onset, its inclusion in clustering may bias results. I continue to recommend conducting a sensitivity analysis that excludes this variable to evaluate its impact on cluster formation and interpretation. This is critical to support the authors' assertion that diagnosis age is integral to their model.

RESPONSE: We agree that there is value in such analysis. We have run the recommended analysis. We document this as a sensitivity analysis in the methods: "Third, to determine if analogous clusters could be derived without age of diagnosis data, we repeated the clustering using binary indicators of disease presence at death." [Lines 493 – 495]

We document the results briefly in the manuscript: "Among the IBD cohort, clustering with binary indicators produce a 2-cluster solution, which appear to lump α-Clusters and β-Cluster into one cluster (Cluster A), whilst retaining the γ-Cluster (**Supplemental Table S13**)." [Lines 202 – 205] We provide further detail in supplemental file. For ease of review, we have also added the supplemental figures and table for this sensitivity analysis to the bottom of this document.

8. The previous comment 6: Please carefully revise the mistakes in the figures. The numbers in Figures 7a and 7b for the original Figures 6), the deviation and validation cohorts are still identical in the manuscript.

RESPONSE: Our apologies. We had updated the figure in the manuscript files but not in the manuscript itself. This new figure is now attached.

9. Minor: "We excluded individuals with an invalid 424 health card, missing age, and/or missing date of death…" Please remove "and/" to avoid logical ambiguity.

RESPONSE: We have removed the "and/or". It now reads: "We excluded individuals with an invalid health card, missing age, or missing date of death" [Lines 347-348]

Reviewer #2 (Remarks to the Author)

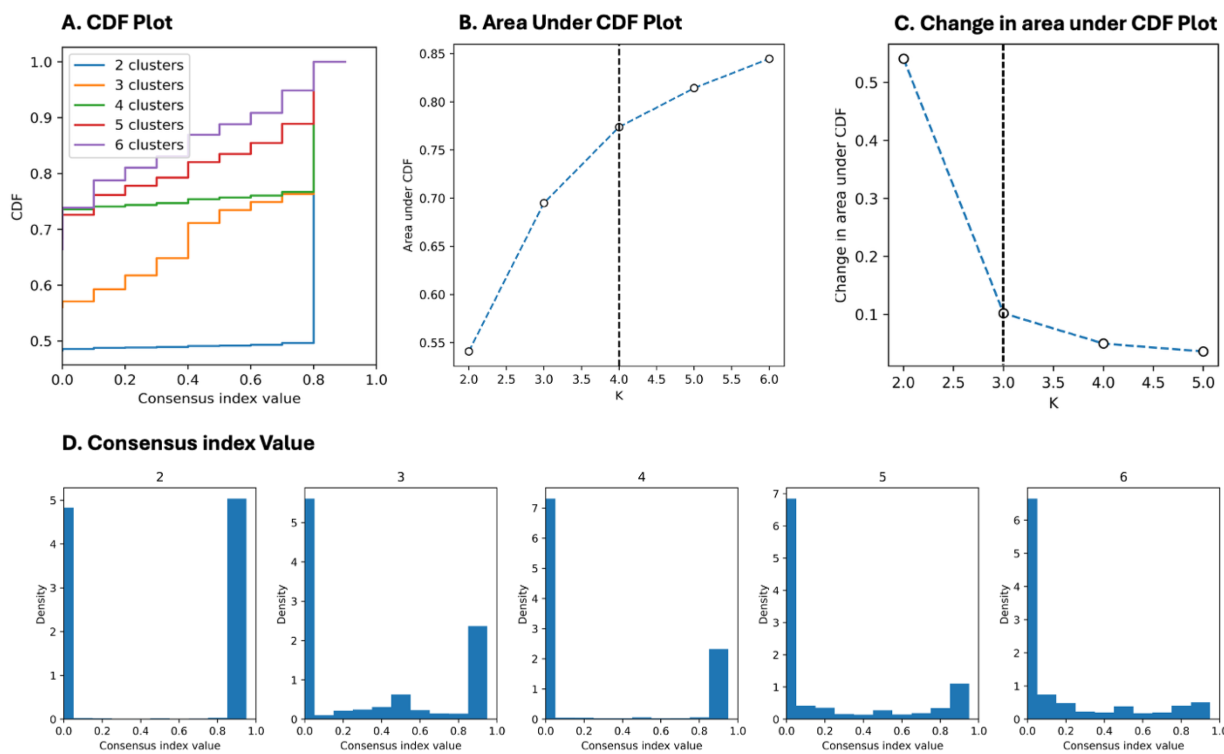
Thank you for your further clarification. I do not have any other comments to add. Great works!

RESPONSE: Thank you for your additional review of the manuscript.

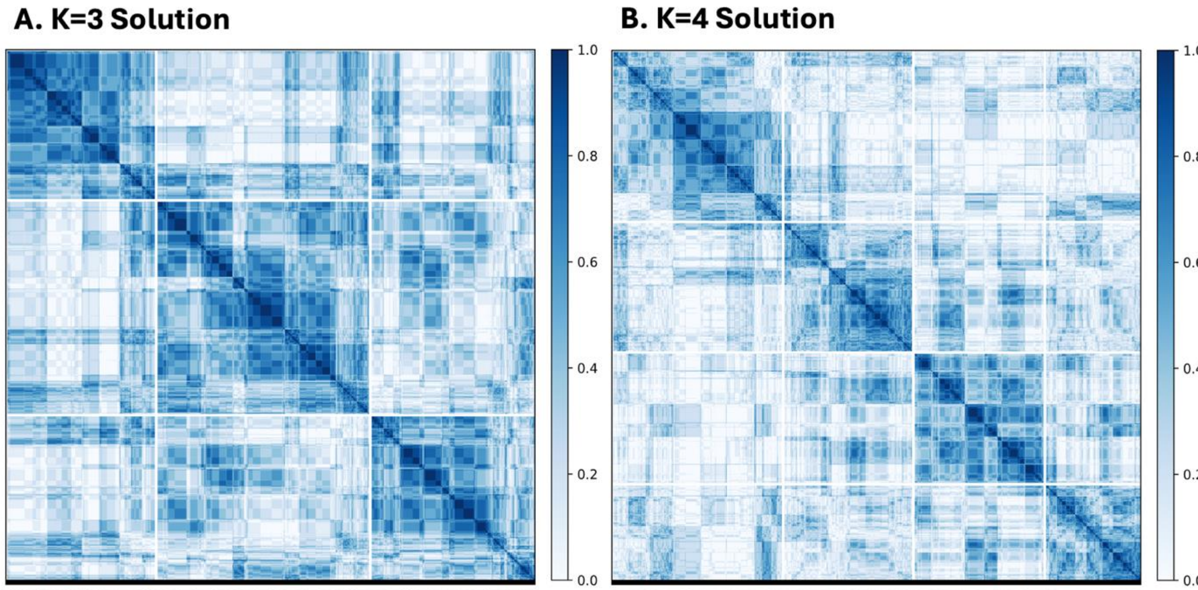
(1) Sensitivity Analysis Evaluating Specificity of Clusters to IBD

Re-Derivation of Clusters on Matched Controls

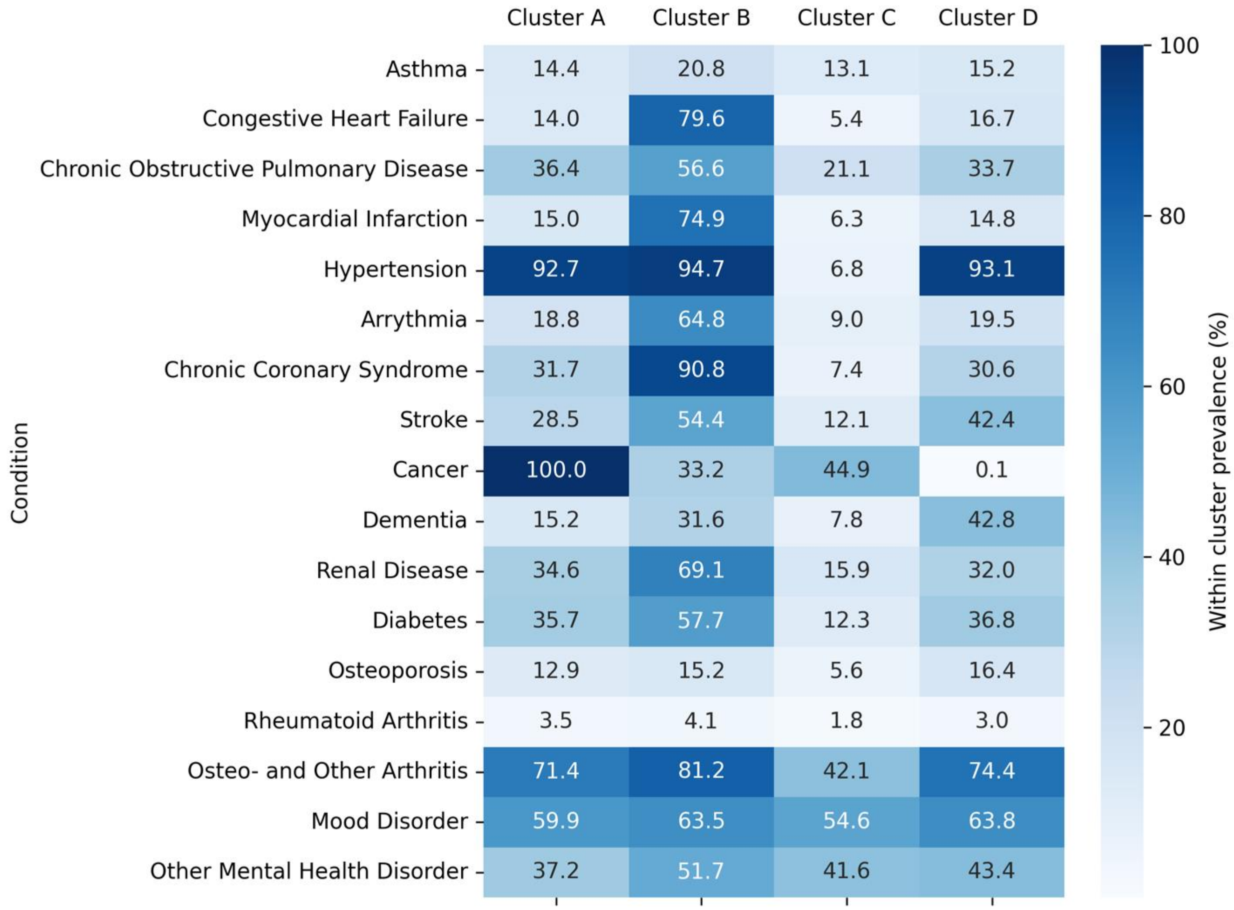
We used analogous methodology to re-derive the clusters in the matched controls. This approach highlighted that, among controls, K=4 was the best clustering solution (vs. K=3 in IBD). We emphasize that these patterns were derived among matched controls (matched on year of birth, year of death, and sex); thus, this analysis is not intended to produce inference about clusters of chronic conditions in the entire general population.



Supplemental Figure S18. Cumulative Density Function (CDF) plots of the clustering solution produced using matched control cohort for K=2 to K=6.



Supplemental Figure S19. Consensus matrices of the K=3 and K=4 clustering solutions produced with the matched control cohort. These solutions reflect the two best solutions identified on the CDF plots. Interpretation: The K=4 solution is superior to the K=3 solution. The absence of sharp blue squares indicates that discovered clusters are not as stable among multiple clustering runs as that produced using the IBD Cohort.



Supplemental Figure S20. Heatmap of condition prevalence across the four clusters identified on the matched controls dataset.

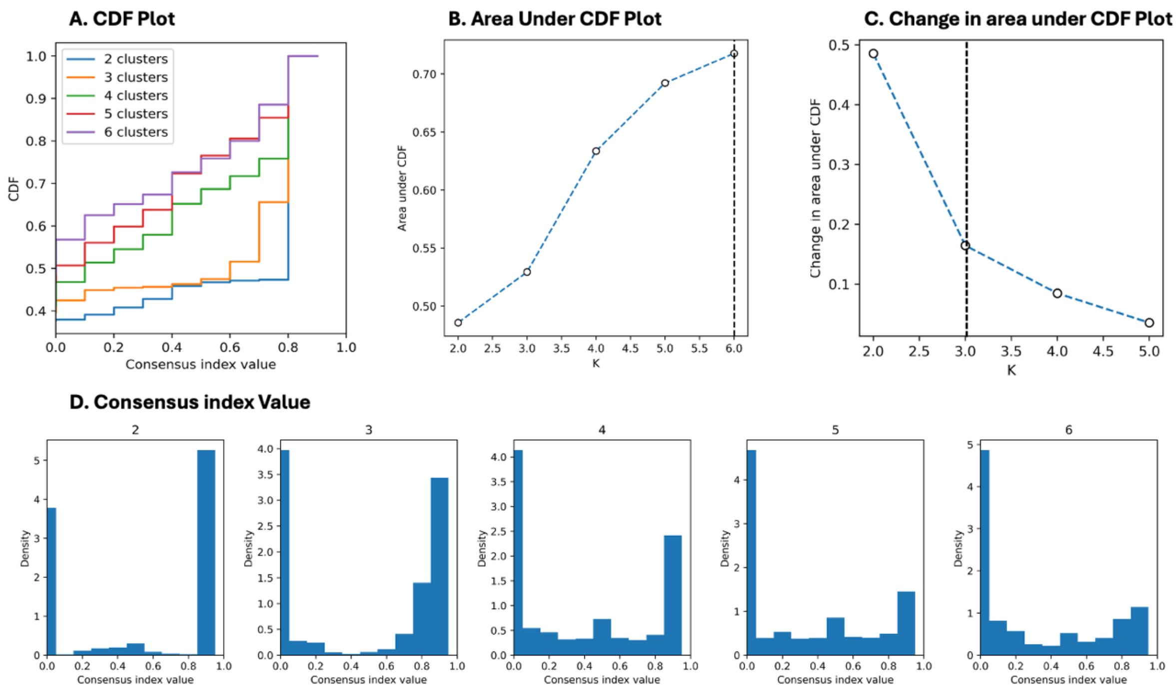
Supplemental Table S12. Description of K=4 clusters derived with the matched controls cohort (N=11,598, 25% random sample)

N(%) or Median [IQR]	Cluster A (N=2836)	Cluster B (N=3160)	Cluster C (N=2876)	Cluster D (N=2726)
Non-Cluster Deriving Variables				
Age of death (years), median [Q1, Q3]	76 [68, 84]	83.0 [75.0,88.0]	59.0 [49.0,69.0]	80.0 [68.0,87.0]
Number of chronic conditions, median [Q1, Q3]	6 [5,7]	9.0 [8.0,11.0]	3.0 [2.0,4.0]	6.0 [5.0,7.0]
Cluster Deriving Variables				
Asthma, n (%)	**408 (14.4)**	**657 (20.8)**	**378 (13.1)**	**414 (15.2)**
Age of Diagnosis (years), median [Q1, Q3]	59 [50, 68]	66 [57, 74]	37 [27, 51]	60 [47, 69]
Congestive Heart Failure, n (%)	**396 (14.0)**	**2,516 (79.6)**	**156 (5.4)**	**455 (16.7)**
Age of Diagnosis (years), median [Q1, Q3]	74 [65, 82]	76 [67, 83]	60 [53, 70]	76 [65, 85]
Chronic Obstructive Pulmonary Disease, n (%)	**1,031 (36.4)**	**1,787 (56.6)**	**607 (21.1)**	**919 (33.7)**
Age of Diagnosis (years), median [Q1, Q3]	65 [57, 73]	70 [61, 79]	55 [47, 63]	66 [56, 75]
Myocardial Infarction, n (%)	**426 (15.0)**	**2,368 (74.9)**	**182 (6.3)**	**403 (14.8)**
Age of Diagnosis (years), median [Q1, Q3]	68 [60, 76]	72 [63, 80]	55 [46, 64]	68 [58, 77]
Hypertension, n (%)	**2,628 (92.7)**	**2,991 (94.7)**	**196 (6.8)**	**2,537 (93.1)**
Age of Diagnosis (years), median [Q1, Q3]	62 [54, 70]	65 [58, 71]	45 [37, 52]	64 [54, 72]
Arrythmia, n (%)	**532 (18.8)**	**2,047 (64.8)**	**259 (9.0)**	**531 (19.5)**
Age of Diagnosis (years), median [Q1, Q3]	71 [63, 79]	73 [65, 80]	58 [43, 67]	73 [64, 81]
Chronic Coronary Syndrome, n (%)	**900 (31.7)**	**2,870 (90.8)**	**213 (7.4)**	**835 (30.6)**
Age of Diagnosis (years), median [Q1, Q3]	65 [58, 73]	68 [61, 76]	56 [45, 64]	66 [58, 74]
Stroke, n (%)	**809 (28.5)**	**1,719 (54.4)**	**348 (12.1)**	**1,155 (42.4)**
Age of Diagnosis (years), median [Q1, Q3]	71 [62, 79]	73 [65, 80]	56 [45, 69]	73 [63, 81]
Cancer, n (%)	**2,836 (100.0)**	**1,049 (33.2)**	**1,292 (44.9)**	**1-5 (≤0.1)**
Age of Diagnosis (years), median [Q1, Q3]	70 [62, 78]	72 [64, 81]	57 [49, 66]	48 [47, 49]
Dementia, n (%)	**430 (15.2)**	**1,000 (31.6)**	**225 (7.8)**	**1,168 (42.8)**
Age of Diagnosis (years), median [Q1, Q3]	80 [75, 85]	82 [76, 87]	73 [65, 82]	80 [73, 85]
Renal Disease, n (%)	**982 (34.6)**	**2,185 (69.1)**	**458 (15.9)**	**872 (32.0)**
Age of Diagnosis (years), median [Q1, Q3]	71 [62, 80]	76 [68, 83]	54 [44, 65]	72 [61, 82]
Diabetes, n (%)	**1,013 (35.7)**	**1,824 (57.7)**	**354 (12.3)**	**1,003 (36.8)**
Age of Diagnosis (years), median [Q1, Q3]	63 [56, 71]	66 [57, 75]	52 [44, 61]	63 [53, 72]
Osteoporosis, n (%)	**365 (12.9)**	**479 (15.2)**	**162 (5.6)**	**447 (16.4)**
Age of Diagnosis (years), median [Q1, Q3]	69 [61, 76]	74 [67, 81]	61 [52, 69]	72 [65, 79]
Rheumatoid Arthritis, n (%)	**98 (3.5)**	**130 (4.1)**	**53 (1.8)**	**83 (3.0)**
Age of Diagnosis (years), median [Q1, Q3]	65 [54, 74]	67 [58, 75]	54 [45, 66]	63 [51, 73]
Osteo- and Other Arthritis, n (%)	**2,025 (71.4)**	**2,567 (81.2)**	**1,211 (42.1)**	**2,028 (74.4)**
Age of Diagnosis (years), median [Q1, Q3]	61 [52, 69]	65 [57, 72]	45 [36, 56]	63 [51, 71]
Mood Disorder, n (%)	**1,699 (59.9)**	**2,008 (63.5)**	**1,569 (54.6)**	**1,738 (63.8)**
Age of Diagnosis (years), median [Q1, Q3]	59 [49, 69]	66 [56, 74]	40 [30, 52]	61 [48, 72.0]
Other Mental Health Disorder, n (%)	**1,056 (37.2)**	**1,635 (51.7)**	**1,197 (41.6)**	**1,182 (43.4)**
Age of Diagnosis (years), median [Q1, Q3]	64 [52, 74]	70 [60, 80]	43 [31, 55]	62 [49, 75]

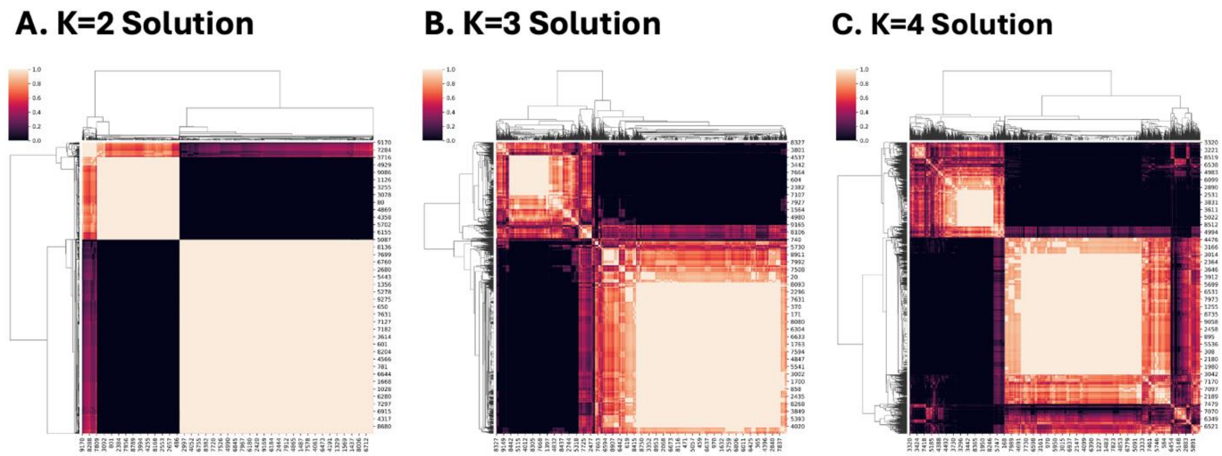
(2) Sensitivity Analysis Evaluating Specificity of Clusters to Age Data

Re-Derivation of Clusters with Binary Disease Indicators

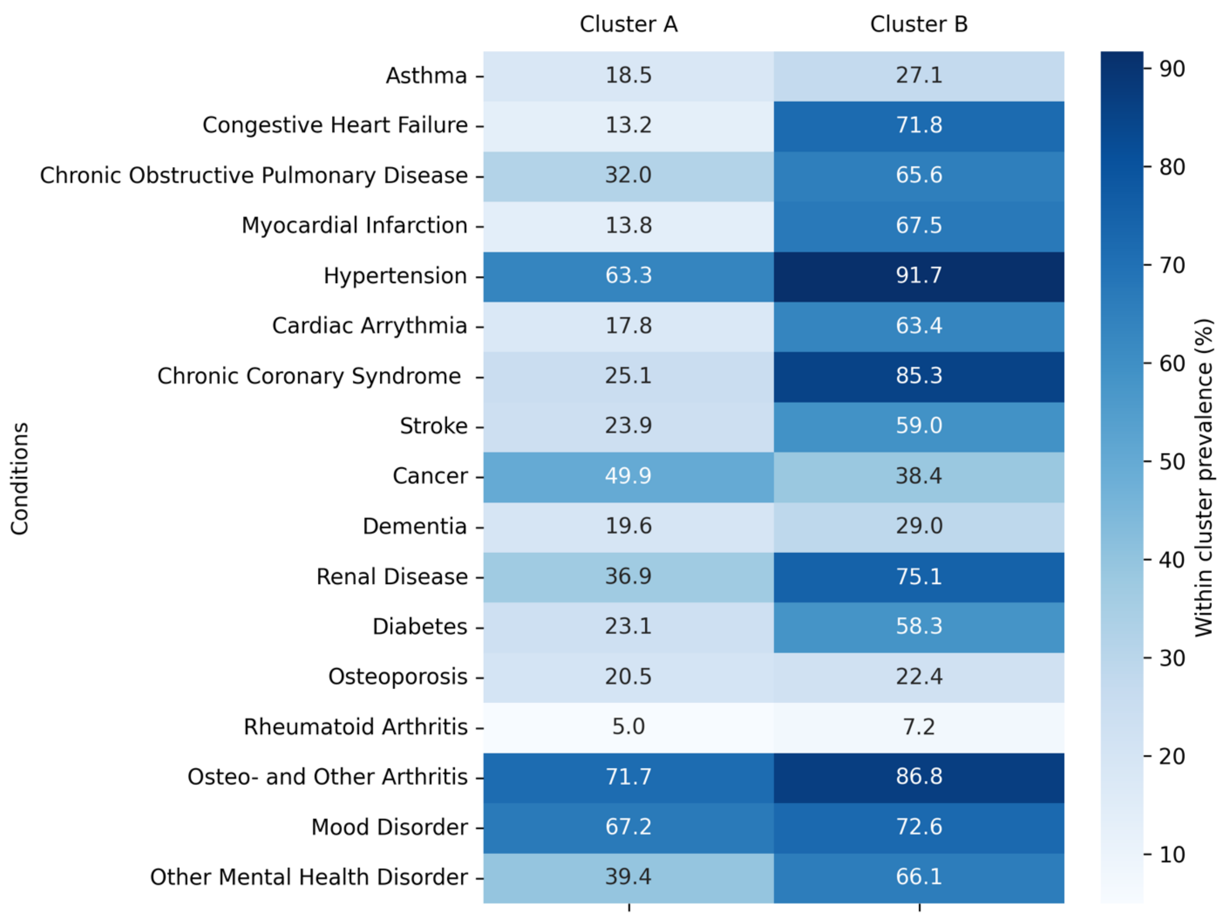
In the primary analysis, clusters were derived using numerical data that reflected the age at which the individual sought care for chronic conditions. To evaluate the specificity of our findings to such data, we re-derived the clusters using binary indicators of disease presence at death. We repeated the initial analysis with K-Modes as the base model of the consensus clustering solution to adequately model binary data. The results demonstrated strong performance for the K=2 and K=3 solutions; however, the size of the third cluster generated in the K=3 solutions were not clinically meaningful.



Supplemental Figure S21. Cumulative Density Function (CDF) plots of the clustering solutions produced with the binary disease indicator data from the IBD cohort for values of *K* between *K*=2 to *K*=6.



Supplemental Figure S22. Consensus matrices of the K=2, K=3, and K=4 clustering solutions produced with the binary disease indicator data from the IBD cohort. These clustering solutions were the best solutions identified on the CDF plots. Interpretation: Sharp light-colored squares indicate stable clusters across among multiple clustering runs. The K=3 and K=4 clustering solutions essentially replicate the K=2 clustering solutions, pulling out clusters that are non-meaningful in size.



Supplemental Figure S23. Heatmap of condition prevalence across the two clusters identified using binary disease indicators on the IBD dataset.